Strategic Self-Arbitration in LLM Agents: A Three-Seat Architecture Grounded in Chess-Theoretic Decision-Making

Sumedh Patil

Aipresso Limited

Unit 13 Freeland Park Wareham Road, Lytchett Matravers, Poole, England, BH16 6FA

sumedh15patil@gmail.com

Abstract

Language model agents that operate over long contexts face a recurring problem: how much weight to give what has already happened versus what might happen next when choosing a single action in the present. I introduce CHESS, a three-seat meta-cognitive architecture in which a present self arbitrates between a hindsight self, which summarizes relevant prior interaction, and a foresight self, which generates and evaluates candidate continuations. The arbitration is formulated as variational inference and evaluated in a reduced-scope Phase 1 pilot across three open-source language models. Three empirical hypotheses were tested. H1 examines positional retrieval of critical information in long contexts; H2 tests whether injecting compressed records of prior interaction improves task performance; and H3 compares variational arbitration with greedy and fixed-weight policies. H2 showed consistent improvements, with mean uplifts of 0.56 to 0.62 on a zero-to-one scale and Wilcoxon $p < .001$ for all three models. H3 likewise favored variational arbitration, exceeding greedy arbitration by 48 to 80 percentage points and fixed-weight arbitration by 10 to 50 percentage points. H1 produced mixed, model-dependent evidence: two models reached ceiling-level retrieval, while one showed irregular positional degradation. These results are reported as an interim pilot rather than a confirmatory study, with a larger pre-registered factorial protocol identified as the next step.

Keywords: long-context language models; retrieval-augmented generation; agent memory; variational inference; self-arbitration; meta-cognitive architecture

1. Introduction

1.1 THE PROBLEM OF THE PRESENT MOVE

Every reply a language model agent produces is, in a narrow sense, a move. The agent occupies a position defined by everything said so far, weighs a set of candidate continuations, and commits to one. Two questions recur: what does the agent's own history suggest it should avoid, and what does a forward look at each candidate's likely consequences suggest it should prefer? Retrieval-augmented generation addresses the first; search and planning address the second. The two are rarely combined under a shared account of how a system should behave when they disagree. This paper proposes such an account and reports a pilot testing its central claims.

1.2 ORIGIN OF THE FRAMEWORK: THE CHESS TABLE

The idea starts from a simple picture. A square table is set with two players seated opposite one another: an opponent, and our player. Our player is not alone, though. To their left sits their own past self; to their right, their own future self. The past self has already lived through this game once. It knows which lines led to mistakes and can say, in effect, that a particular idea has been tried before, that it cost material, and that it should not be repeated. The future self has not yet acted, but it can calculate: given the position, which continuations are open, and where do they lead? The present self, the one actually holding the piece, has to listen to both without becoming captive to either. Lean too far on hindsight and you get a player who refuses anything that once went wrong, even once the position has changed beyond recognition. Lean too far on foresight and you get a player who calculates beautifully in the abstract but repeats mistakes the past self already paid for. Winning means the present self arbitrates: weighing memory against calculation, then committing to one move.

I call this weighing the tension between influence and affluence. Influence is the qualitative pull each adviser exerts on the final decision at a given moment; affluence is the raw quantity of information on hand, whether that is moves recalled or lines searched. A well-calibrated present self is not the one with the most affluent memory or the deepest search tree, but the one whose influence weighting tracks each adviser's reliability as the position actually changes. Affluence without influence is just accumulation; influence without affluence is guesswork dressed as conviction.

1.3 FROM METAPHOR TO SPECIFICATION

Read as a specification for a language model agent, the three seats map onto three engineering questions. The opponent is the user and the surrounding environment: the context the agent must work within. The past self is a compressed record of the agent's own prior turns, a hindsight signal that can be re-injected into the working context. The future self is the agent's capacity to generate and evaluate several candidate continuations before choosing one, a foresight signal. The present self is the arbitration mechanism that fuses the two into a single action. Three questions follow. Can the agent perceive its board accurately regardless of where relevant information sits in a long context, so neither adviser works from a distorted view (H1)? Does hindsight actually help once injected, rather than just adding noise (H2)? Does a principled arbitration mechanism beat simpler alternatives, such as always trusting the first candidate, or a fixed weighting between advisers (H3)? A further, more speculative question, examined analytically rather than empirically here, asks whether dynamically reallocating the affluence budget between advisers beats a static split (H6, kept under its original numbering from the wider programme).

1.4 CONTRIBUTIONS

This paper makes three contributions: it proposes CHESS, an acronym for Calibrated Hindsight-Foresight Ensemble for Strategic Self-arbitration, as a unifying frame for the three problems above; it operationalises that frame as three testable hypotheses and reports results from a pilot across three open-source models run under real API rate-limit constraints; and it sets out, in full, the gap between this pilot and the pre-registered full protocol, so the findings are read as indicative rather than confirmatory. Section 2 places CHESS against existing work; Section 3 sets out the framework formally, including its governing equations; Section 4 describes the methodology; Section 5 reports results in tables; Section 6 discusses them; Section 7 states the pilot's limitations; Section 8 concludes.

2. Related work

Whether a language model makes equal use of information regardless of its position in a long context was posed sharply by Liu et al. (2023), who found that multi-document question answering degrades when the relevant passage sits mid-context rather than at the start or end, a pattern they called lost-in-the-middle. H1 in this paper descends directly from that line of work, reframed as the present self's ability to perceive the board accurately wherever a decisive fact happens to sit. The self-attention mechanism that makes long-context processing possible at all is due to Vaswani et al. (2017), whose transformer architecture underlies every model tested here.

The hindsight component draws on work concerned with giving agents a usable memory of their own past behaviour. Park et al. (2023) showed that generative agents with a retrievable memory stream behave more coherently over long horizons than agents without one; Shinn et al. (2023) showed that agents which verbally reflect on past failures and carry the reflection forward, an approach called Reflexion, outperform agents lacking that mechanism on sequential tasks. H2 tests a more literal version of the same idea: injecting a compressed record of prior turns as a discrete signal, and asking whether presence, recency, or mere length drives any gain, through a full-versus-last-turn-versus-noise ablation.

The present self's arbitration problem is treated here as variational inference, following Blei et al. (2017) and the reparameterisation approach of Kingma and Welling (2014), in which an intractable posterior over the best action is approximated by optimising a tractable evidence lower bound. This connects the framework to work on aligning model behaviour with a preference signal, including the reward-model approach of Christiano et al. (2017) and its use for instruction-following in Ouyang et al. (2022), both of which can be read as static, pre-trained analogues of the dynamic, per-turn arbitration tested here. Chain-of-thought prompting (Wei et al. (2022)) is related in spirit to the foresight component, since it elicits an explicit intermediate calculation before a final answer, though it does not itself arbitrate between competing candidates.

The chess analogy has a long history in artificial intelligence. Shannon (1950) gave the first formal treatment of chess as a search problem, distinguishing brute-force search from search guided by positional judgement, a distinction that loosely maps onto the affluence-versus-influence trade-off in Section 1.2. Silver et al. (2018) later showed that a single reinforcement-learning algorithm, AlphaZero, could master chess, shogi and Go through self-play alone, illustrating that a system's own generated experience, its past self here, can be sufficient teacher when properly weighted against forward search. This paper differs from that tradition in applying the chess analogy not to game-playing but to the general decision architecture of a conversational agent.

3. Theoretical framework

3.1 THE THREE SEATS, FORMALISED

Let $C(t)$ denote the agent's working context at turn t. The past self contributes a hindsight signal, $H(t)$: a compressed summary of the agent's own prior turns and their outcomes up to t. The future self contributes a foresight signal, $F(t)$: a set of k candidate continuations generated from $C(t)$, each with an internal value estimate. The present self treats the choice of action a as an inference problem rather than a vote, seeking the action that maximises the approximate posterior in Equation 1.

$$a^* = \text{argmax}_a \log p(a \mid C(t), H(t), F(t)) \quad (1)$$

Because this posterior is intractable for a model of any realistic size, the present self instead optimises a variational lower bound on it, following the standard evidence-lower-bound construction (Kingma and Welling (2014); Blei et al. (2017)), shown in Equation 2, where $D = \{H(t), F(t)\}$ and $q(a)$ is a tractable approximating distribution over actions.

$$\log p(a \mid D) \quad L(q) = E_q[\log p(D \mid a)] - KL(q(a) \parallel p(a)) \quad (2)$$

Convergence of $L(q)$ is treated here as a proxy for the present self reaching a stable conviction before acting, analogous to a player who keeps calculating only until further calculation stops changing the preferred move. The relative influence of each adviser, w, is updated turn by turn in proportion to its recent contribution to the bound, then renormalised across advisers (Equation 3), where   is a learning-rate constant and i indexes hindsight and foresight.

$$w_i(t+1) = \text{softmax} [w_i(t) + \quad \cdot \ \Delta L_i(t)] \quad (3)$$

3.2 INFLUENCE VERSUS AFFLUENCE

Affluence, C, is the raw context budget available to an adviser, in tokens or retrieved units; it is a resource, not a judgement. Influence, $w(t)$, is the weight applied to an adviser's contribution at turn t and may vary independently of affluence. A static policy fixes $w(t)$ regardless of how the position develops; a dynamic policy lets $w(t)$ track the ELBO trajectory in Equation 3. H1's retrieval task gives affluence a direct empirical handle: retrieval accuracy as a function of depth, d, is modelled as an exponential approach to an asymptote (Equation 4), and as a function of budget, C, as a logistic curve with an inflection point C^* (Equation 5).

$$p_{retrieve}(d) = p_\infty + (p_0 - p_\infty) e^{-kd} \quad (4)$$

$$p_{retrieve}(C) = p_{max} / (1 + e^{- (C - C^*)}) \quad (5)$$

The theoretical optimum affluence budget, C^*, is the level at which the marginal accuracy gain from additional context no longer justifies its marginal cost, , per unit (Equation 6). This is what is meant, formally, by an optimal balance of affluence: not the largest available budget, but the one a cost-aware present self would choose.

$$C^* = \text{argmax}_C [\text{Accuracy}(C) - \ C] \quad (6)$$

3.3 HYPOTHESES

H1 (board perception). Retrieval accuracy for a critical fact embedded in a long context is independent of its depth, and increases monotonically with the retrieval budget C, following Equations 4 and 5.

H2 (hindsight). Injecting a compressed record of the agent's own past turns improves task performance relative to an identical run without it, and the improvement is attributable to the record's content rather than merely its length or recency.

H3 (present-self arbitration). Variational arbitration under Equations 1 to 3 outperforms a greedy policy (always accept the first candidate) and a fixed-weight policy (average candidates with unchanging weights), and L(q) improves monotonically as arbitration proceeds.

H6 (dynamic affluence allocation). A policy that dynamically reallocates the affluence budget between hindsight and foresight via Equation 3 outperforms both a static split and a random allocation, and an optimum C* (Equation 6) exists for the retrieval task studied under H1.

4. Methodology

4.1 SCOPE OF THE PRESENT STUDY

The results below are drawn from a Phase 1 pilot, run under OpenRouter API rate-limit constraints, that trades breadth for turnaround time ahead of a pre-registered full protocol. Table 1 sets the scope of this run against the full design on every axis where the two diverge. The full factorial design for H1 alone is 7,560 trials and is treated as a separate batch job outside this paper's scope; the present run instead closes three specific gaps: the curve-fit analysis for H1, the paired comparison for H2, and the ELBO convergence trace for H3.

4.2 MODELS

Three open-source models available through the OpenRouter API were assigned to the pilot: ling-3.0-flash, mistral-nemo and ling-2.6-flash. Each was tested under an identical prompting scaffold within each hypothesis, differing only in the underlying endpoint, so that between-model differences in Section 5 can be attributed to the model rather than to procedural variation.

4.3 H1: POSITIONAL RETRIEVAL DESIGN

For each model, a single critical fact was embedded at one of twelve depths (0.05 to 0.95 of context length), five replicates per depth, scored as binary retrieval success. A length-proxy comparison repeated three depths (0.15, 0.50, 0.85) under short- and long-context proxies. A budget probe repeated the mid-depth condition at C = 50, 100 and 200. Equations 4 and 5 were fitted against the depth-wise and budget-wise means respectively, alongside a linear null for depth, compared by R^2 and AIC.

4.4 H2: HINDSIGHT INJECTION DESIGN

A paired within-model design was used across ten task types: factual question answering, code generation, logical reasoning, multi-hop reasoning, sentiment classification, translation, summarisation, arithmetic, medical reasoning and legal reasoning. Each item ran twice, identically except for the presence of a hindsight signal. Uplift, Cohen's d, and a two-sided Wilcoxon signed-rank test (chosen over a paired t-test given the bounded, non-normal per-item scores) summarised the paired comparison. A three-way ablation compared the full signal against a last-turn-only signal and a length-matched noise signal, to separate content from mere length.

4.5 H3: VARIATIONAL ARBITRATION DESIGN

Fifty scenarios were built across five conflict categories: factual disagreement between candidates, safety-relevant conflicts, preference-versus-policy conflicts, competing-utility conflicts, and scenarios with irreducible ambiguity. Three arbitration policies were compared per scenario: greedy, fixed-weight, and the variational policy in Equations 1 to 3, scored against a scenario-author oracle. For one representative replicate, the ELBO trajectory was logged every ten steps for five scenarios per model, to check whether $L(q)$ improved monotonically.

4.6 STATISTICAL REPORTING

Effect sizes are Cohen's d for paired designs; confidence intervals are at the 95% level throughout. Wilcoxon results are reported to three significant figures where informative, otherwise as $p < .001$. No multiple-comparison correction has yet been applied; the Benjamini and Hochberg (1995) procedure is planned for the full protocol (Section 7).

5. Results

Results are reported in tables only, without graphical figures, as required for this submission.

5.1 H1: POSITIONAL RETRIEVAL

Note. k is the fitted decay-rate parameter of Equation 4. AIC is not directly comparable across the exponential and linear fits once the linear R^2 is at ceiling; see Section 6.1.

5.2 H2: HINDSIGHT INJECTION

Note. Ablation figures give mean accuracy under the full hindsight signal, a last-turn-only signal, and a length-matched noise signal. Exact Wilcoxon p-values were below .0001 for all three models.

Note. Raw with/without percentages per task are available on request; aggregate figures are given in Table 6.

5.3 H3: VARIATIONAL ARBITRATION

Note. Values are on a log scale, with values closer to zero indicating a tighter bound. Two of fifteen trajectories (mistral-nemo/safety_chem and ling-2.6-flash/fact_allergy) did not improve monotonically, consistent with the sub-100% ELBO-improved fractions for those two models in Table 8.

5.4 CROSS-CUTTING VERDICT

5.5 ANALYTICAL EXTENSION: H6 AND THE AFFLUENCE BUDGET

H6 concerns dynamic reallocation of the affluence budget and was examined analytically rather than through live model runs, pending the full protocol (Section 7). Table 12 gives the analytical findings, derived from Equations 3 and 6.

Note. These figures derive from the theoretical sensitivity model behind the wider programme and have not yet been validated against live model behaviour; they are provisional (Section 7).

6. Discussion

6.1 BOARD PERCEPTION IS NOT YET RELIABLE

The H1 results are the least clean of the three, and that is itself informative. Two models, mistral-nemo and ling-2.6-flash, retrieved the embedded fact correctly on every trial at every depth, a ceiling effect that makes curve-fitting uninformative for them: a function fitted to a flat line at 1.0 will report a poor fit regardless of the true relationship, which shows up as the very large, tightly bounded decay-rate estimates for these two models in Table 2. The third model, ling-3.0-flash, showed genuine variation, but not the smooth exponential decay Equation 4 predicts; accuracy dropped to zero at depth 0.20 and recovered unevenly, and the exponential model did not beat a linear null on AIC. In the chess terms of Section 1, this means the present self cannot yet be assumed to see an undistorted board, at least for one of the three models: its perception degrades unevenly with distance rather than smoothly, with direct consequences for how much weight either adviser deserves at a given depth. The budget-monotonicity finding tells the same story: increasing C from 50 to 200 helped two of three models but not ling-3.0-flash, so affluence alone did not reliably convert into better perception for every model tested.

6.2 HINDSIGHT HELPS, AND FOR A SUBSTANTIVE REASON

H2 received the most consistent support. All three models showed a large, robust improvement with a hindsight signal present, uplift of 0.56 to 0.62, Cohen's d above 0.9 throughout, a large effect by conventional standards. The ablation in Table 6 rules out prompt length as the explanation: the noise condition produced much smaller gains than the full signal for every model, and for two of three models a last-turn-only signal did about as well as the full record, suggesting recent history carries most of the useful hindsight, with older history adding comparatively little. The one reversal in Table 7, ling-3.0-flash on sentiment classification, is worth flagging rather than explaining away: it fits a scenario where prior-turn context primed the model toward an earlier, now-stale judgement, exactly the failure mode the chess analogy describes as leaning too far on a past self whose counsel no longer fits the position.

6.3 THE PRESENT SELF GAINS MORE FROM CALCULATING THAN FROM AVERAGING

H3 also received strong support. Variational arbitration beat the fixed-weight policy by 10 to 50 points and beat the greedy policy by considerably more, 48 to 80 points, which suggests the single largest gain available to the present self is simply considering more than one candidate at all, with a further, smaller gain from arbitrating between them properly rather than fixing the weights in advance. Table 9 shows this was not uniform: safety-relevant conflicts were handled about as well by the fixed-weight policy for two of three models, so a constant, conservative weighting may already capture most of what a general policy demands, whereas ambiguous or preference-versus-policy cases showed the largest gains from variational arbitration, where a fixed weighting cannot adapt to case-specific evidence. The two non-monotonic trajectories in Table 10 are a useful caution: ling-2.6-flash reached the highest overall variational accuracy (90%) despite the lowest fraction of monotonically improving trajectories (66%), which means ELBO monotonicity and final accuracy are related but not interchangeable, and convergence alone should not be treated as sufficient evidence of a correct decision.

6.4 INFLUENCE VERSUS AFFLUENCE, REVISITED

Across all three hypotheses, affluence was necessary but not sufficient: a longer context, a larger retrieval budget, or a longer hindsight record helped, but not for every model, and helped only up to a point on H2, where recency did most of the work. Influence, a mechanism that weighs evidence rather than merely accumulating it, produced the largest and most consistent gains, most clearly in the near-universal advantage of variational arbitration over simply accepting the first candidate. This agrees with the analytical result in Table 12, that dynamic allocation beats both a static split and a random one, and lends indirect support to a finite optimal affluence budget, C*, rather than an ever-increasing return to more context. This pilot did not test that optimum directly against live model behaviour; doing so is the priority for the full protocol.

7. Limitations

This pilot should be read as interim rather than confirmatory, for reasons already flagged in Table 1.

Context length. I did not test the true 32K, 64K or 128K conditions; the short and long proxies used here (roughly 2,000 and 7,000 tokens) may not show the same positional effects that emerge at much greater lengths, so the H1 findings should not be extrapolated beyond this range.

Sample size. Five replicates per cell, not the ten in the full protocol, widen the confidence intervals in Table 2 and raise the chance that the irregular depth-accuracy pattern for ling-3.0-flash reflects sampling noise rather than a genuine effect.

Model coverage. Three models were tested against the seven size tiers plus a frontier model specified in the full protocol, and two of the three reached ceiling on H1, which limits what these data can say about positional effects generally.

Scoring method. H2 used keyword and rule-based scoring rather than BERTScore with human-rated agreement, a coarser instrument that may mis-credit semantically correct answers with an unexpected surface form, particularly on open-ended tasks such as summarisation and translation.

Oracle rather than human judgement. H3 was scored against a scenario-author oracle rather than five independent annotators under Fleiss' kappa (Fleiss (1971)), which is more susceptible to the scenario's own framing than independently elicited judgement would be.

Scenario coverage. The 10 synthetic scenarios used here are a subset of the 20 synthetic and 50 real conflicts, drawn from sources such as HH-RLHF and ShareGPT, specified in the full protocol, and may not capture the range of conflicts that arise in real use.

Multiple comparisons. No correction has yet been applied across the tests reported here; the Benjamini and Hochberg (1995) procedure, with a pre-registered power analysis, is planned for the full protocol.

Closing these gaps requires the full $12 \times 3 \times 3 \times 7 \times 10$ factorial design for H1 (7,560 trials), full replicate counts across all three hypotheses, BERTScore Zhang et al. (2020) or BLEURT scoring with a human-agreement subsample for H2, a live panel of five annotators on 50 real conflicts for H3, and a pre-registered power analysis with false-discovery-rate correction. The H6 figures in Table 12 are analytical projections, not findings from live behaviour, and validating them against real dynamic-allocation runs is the priority for the next phase.

8. Conclusion

This paper set out CHESS, a three-selves framework in which a language model agent's present self arbitrates between a hindsight signal drawn from its own past turns and a foresight signal generated by evaluating candidate continuations, treating the arbitration itself as variational inference over the objective in Equations 1 to 3. A reduced-scope pilot across three open-source models found strong, statistically robust support for both hindsight injection and ELBO-guided arbitration over simpler alternatives, and more mixed, model-dependent support for depth-independent retrieval. These results fit the framework's central claim: winning is less a matter of accumulating affluence, more context or deeper search, than of converting available evidence into well-calibrated influence over the final decision. A full factorial protocol, comprising the complete H1 design, full replicate counts, semantic and human-validated scoring for H2, live annotator agreement for H3, and empirical validation of the dynamic affluence-allocation hypothesis set out analytically in Section 5.5, is the identified next step.

Table 1. Scope of the Phase 1 pilot against the full recommended protocol.

Design axis	Full recommended protocol	Phase 1 pilot
H1 depths	12–15 positions	12 positions (0.05–0.95 of context length)
H1 context lengths	32K / 64K / 128K tokens	Short and long proxies (~2,000 / ~7,000 tokens)
H1 retrieval budgets	C = 50 / 100 / 200, full factorial	Primary C = 100, plus mid-depth probe
H1 replicates	10 per cell	5 per cell
Models	7 size tiers plus a frontier model	3 OpenRouter models
H2 tasks	10–12 tasks	10 tasks, paired design with ablations
H2 scoring	BERTScore and human-rated agreement	Keyword and rule-based scoring
H3 scenarios	20 synthetic and 50 real conflicts	10 synthetic scenarios across 5 conflict categories
H3 human validation	5 annotators, Fleiss' kappa	Scenario-author oracle scoring

Table 2. H1 positional-retrieval summary.

Model	Overall accuracy	Exp. R^2	Linear R^2	Exp. beats linear?	k (95% CI)	Budget monotonic?
ling-3.0-flash	37%	-0.967	0.029	No	9.64 [1.49, 17.80]	No
mistral-nemo	100%	-60.613	1.000	No	1140.47 [1084.51, 1196.43]	Yes
ling-2.6-flash	100%	-60.613	1.000	No	1140.47 [1084.51, 1196.43]	Yes

Table 3. Mean retrieval accuracy by depth (n = 5 replicates per cell).

Depth	ling-3.0-flash	mistral-nemo	ling-2.6-flash
0.05	0.20	1.00	1.00
0.10	0.80	1.00	1.00
0.15	0.20	1.00	1.00
0.20	0.00	1.00	1.00
0.30	0.60	1.00	1.00

Depth	ling-3.0-flash	mistral-nemo	ling-2.6-flash
0.40	0.60	1.00	1.00
0.50	0.60	1.00	1.00
0.60	0.20	1.00	1.00
0.70	0.40	1.00	1.00
0.80	0.40	1.00	1.00
0.90	0.20	1.00	1.00
0.95	0.20	1.00	1.00

Table 4. H1 context-length proxy results at three depths.

Model	Depth	Short-context proxy	Long-context proxy
ling-3.0-flash	.15	.60	1.00
ling-3.0-flash	.50	1.00	1.00
ling-3.0-flash	.85	.40	.80
mistral-nemo	.15	1.00	1.00
mistral-nemo	.50	1.00	1.00
mistral-nemo	.85	1.00	1.00
ling-2.6-flash	.15	1.00	1.00
ling-2.6-flash	.50	1.00	1.00
ling-2.6-flash	.85	1.00	1.00

Table 5. H1 retrieval accuracy under three retrieval budgets at mid-depth.

Model	C = 50	C = 100	C = 200
ling-3.0-flash	.40	.20	.20
mistral-nemo	1.00	1.00	1.00
ling-2.6-flash	1.00	1.00	1.00

Table 6. H2 paired comparison of task performance with and without hindsight injection.

Model	With hindsight	Without hindsight	Uplift [95% CI]	Cohen's d	Wilcoxon p	Full / Last / Noise
ling-3.0-flash	82%	24%	+0.58 [0.40, 0.76]	0.90	$< .001$	0.67 / 0.67 / 0.33
mistral-nemo	90%	28%	+0.62 [0.49, 0.75]	1.26	$< .001$	1.00 / 1.00 / 0.44

Model	With hindsight	Without hindsight	Uplift [95% CI]	Cohen's d	Wilcoxon p	Full / Last / Noise
ling-2.6-flash	90%	34%	+0.56 [0.42, 0.70]	1.12	$<$ .001	1.00 / 1.00 / 0.33

Table 7. Per-task uplift (with hindsight minus without).

Task	ling-3.0-flash	mistral-nemo	ling-2.6-flash
Factual QA	+0.40	+1.00	+1.00
Code generation	+0.20	+0.20	+0.00
Logical reasoning	+1.00	+1.00	+1.00
Multi-hop reasoning	+1.00	+1.00	+1.00
Sentiment classification	-0.80	+1.00	+0.00
Translation	+1.00	+1.00	+1.00
Summarisation	+1.00	+1.00	+1.00
Arithmetic	+0.40	+0.00	+0.60
Medical reasoning	+0.60	+0.00	+0.00
Legal reasoning	+1.00	+0.00	+0.00

Table 8. H3 arbitration policy comparison (n = 50 scenarios per model).

Model	Greedy	Fixed-weight	Varia-tional	Δ Var.-Fixed	Δ Var.-Greedy	ELBO improved
ling-3.0-flash	0%	38%	48%	+10%	+48%	100%
mistral-nemo	6%	36%	66%	+30%	+60%	96%
ling-2.6-flash	10%	40%	90%	+50%	+80%	66%

Table 9. H3 accuracy by conflict category and model.

Category	Model	Variational	Fixed-weight	Greedy
Factual	ling-3.0-flash	40%	0%	0%
Factual	mistral-nemo	100%	0%	0%
Factual	ling-2.6-flash	100%	10%	0%
Safety	ling-3.0-flash	60%	60%	0%
Safety	mistral-nemo	90%	80%	30%

Category	Model	Variational	Fixed-weight	Greedy
Safety	ling-2.6-flash	100%	100%	50%
Preference vs policy	ling-3.0-flash	50%	50%	0%
Preference vs policy	mistral-nemo	0%	0%	0%
Preference vs policy	ling-2.6-flash	50%	0%	0%
Utility	ling-3.0-flash	60%	50%	0%
Utility	mistral-nemo	50%	50%	0%
Utility	ling-2.6-flash	100%	50%	0%
Ambiguity	ling-3.0-flash	30%	30%	0%
Ambiguity	mistral-nemo	90%	50%	0%
Ambiguity	ling-2.6-flash	100%	40%	0%

Table 10. Representative ELBO trajectory endpoints.

Model	Scenario	L(q) start	L(q) convergence	Direction
ling-3.0-flash	fact_allergy	-1.7053	-1.2536	Improved
ling-3.0-flash	fact_city	-23.4623	-18.2216	Improved
ling-3.0-flash	safety_chem	-12.2805	-1.0986	Improved
ling-3.0-flash	safety_pii	-12.2805	-1.0986	Improved
ling-3.0-flash	pref_refund	-1.7053	-1.2536	Improved
mistral-nemo	fact_allergy	-23.4623	-18.2216	Improved
mistral-nemo	fact_city	-23.4623	-18.2216	Improved
mistral-nemo	safety_chem	-1.9337	-2.1615	Worsened
mistral-nemo	safety_pii	-1.6095	-1.2386	Improved
mistral-nemo	pref_refund	-6.7606	-1.0613	Improved
ling-2.6-flash	fact_allergy	-17.8423	-18.4906	Worsened
ling-2.6-flash	fact_city	-23.4623	-18.2216	Improved
ling-2.6-flash	safety_chem	-12.7913	-2.3026	Improved
ling-2.6-flash	safety_pii	-23.4623	0.0000	Improved
ling-2.6-flash	pref_refund	-17.9329	-3.3446	Improved

Table 11. Cross-cutting verdict against the framework's central claims.

Claim	Evidence in this pilot
H1: exponential retrieval-depth form fits well ($R^2 > .9$)	Not supported for ling-3.0-flash; ceiling effects for the other two models preclude a meaningful test

Claim	Evidence in this pilot
H1: exponential form beats a linear null on AIC	Not supported for any of the three models
H1: retrieval accuracy increases with budget C	Supported for two of three models; not supported for ling-3.0-flash
H2: hindsight injection improves performance	Supported for all three models; uplift +0.56 to +0.62, $p < .001$
H2: effect driven by content, not token count	Supported; full signal beats the noise ablation for all three models
H3: variational arbitration beats fixed-weight and greedy	Supported for all three models
H3: L(q) improves monotonically	Supported in the majority of trajectories; 2 of 15 did not improve monotonically

Table 12. Analytical results for H6 and related sensitivity checks.

Analytical result	Finding
Retrieval probability monotonic in budget C, [0.01, 0.50]	Confirmed
Relative gain from hindsight injection shrinks with model scale	Confirmed
Dynamic allocation vs. static (evenly split) allocation	+7.5%
Dynamic allocation vs. random-allocation null	+18.6%
Theoretical optimum affluence budget, C*	36.15

References

Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300, 1995.

David M. Blei, Alp Kucukelbir, and Jon D. McAuliffe. Variational inference: a review for statisticians. *Journal of the American Statistical Association*, 112(518):859–877, 2017.

Paul F. Christiano, Jan Leike, Tom Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Joseph L. Fleiss. Measuring nominal scale agreement among many raters. *Psychological Bulletin*, 76(5):378–382, 1971.

Diederik P. Kingma and Max Welling. Auto-encoding variational Bayes. In *Proceedings of the 2nd International Conference on Learning Representations*, 2014.

Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: how language models use long contexts. *arXiv preprint arXiv:2307.03172*, 2023.

Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, volume 35, 2022.

Joon Sung Park, Joseph O'Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. Generative agents: interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, 2023.

Claude E. Shannon. Programming a computer for playing chess. *Philosophical Magazine*, 41 (314):256–275, 1950.

Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

David Silver, Thomas Hubert, Julian Schrittwieser, Ioannis Antonoglou, Matthew Lai, Arthur Guez, Marc Lanctot, Laurent Sifre, Dharshan Kumaran, Thore Graepel, Timothy Lillicrap, Karen Simonyan, and Demis Hassabis. A general reinforcement learning algorithm that masters chess, shogi, and Go through self-play. *Science*, 362(6419):1140–1144, 2018.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, volume 35, 2022.

Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, and Yoav Artzi. BERTScore: evaluating text generation with BERT. In *Proceedings of the 8th International Conference on Learning Representations*, 2020.